EXPERIMENT 27: ANTIHISTAMINES – PROTECTION AGAINST ANAPHYLACTIC SHOCK

40 STANDARD IN-COURSE MCQs

1. The main objective of Experiment 27 is to demonstrate:

A. The sedative efficacy of general anesthetics B. Protection provided by antihistamine drugs against anaphylaxis and histamine actions C. The neuromuscular blocking potency of curare D. The rate of gastric emptying in guinea pigs

Answer: B

2. What experimental animal species is designated in the protocol on Page 136?

A. Albino mice B. Wistar rats C. Guinea pigs (200–300 g) D. New Zealand white rabbits

Answer: C

3. Why are guinea pigs the classic animal model for studying histamine and anaphylactic bronchospasm?

A. They have no histamine receptors in the gut B. Their bronchiolar smooth muscle is exquisitely sensitive to histamine and contains abundant H_1 receptors C. They lack mast cells entirely D. They do not produce immunoglobulin antibodies

Answer: B

4. What apparatus is used to expose the animals to aerosolized antigen or histamine?

A. Metabolic cage B. Perspex cage fitted with a spraying/nebulizer unit C. Glass beaker covered with gauze D. Water bath chamber

Answer: B

5. What is the concentration of the adrenaline solution specified on Page 136?

A. 1:100 (10 mg/mL) B. 1:1,000 (1 mg/mL) C. 1:10,000 (0.1 mg/mL) D. 1:100,000 (0.01 mg/mL)

Answer: B

6. By which route is adrenaline administered in the experimental protocol?

A. Oral (p.o.) B. Subcutaneous (s.c.) C. Intraperitoneal (i.p.) D. Intravenous (i.v.)

Answer: B

7. What volume of 1:1,000 adrenaline solution is administered to the guinea pig?

A. 0.05 mL B. 0.1 mL C. 1.0 mL D. 5.0 mL

Answer: C

8. Which H_1 antihistamine is specifically listed on Page 136 with a dose of 50–100 mg/kg?

A. Chlorpheniramine B. Mepyramine C. Cetirizine D. Loratadine

Answer: B

9. How long is allowed after mepyramine injection before aerosol exposure?

A. 2 minutes B. 10 minutes C. 30–45 minutes D. 2 hours

Answer: C

10. How long is allowed after adrenaline subcutaneous injection before aerosol exposure?

A. 2 minutes B. 15 minutes C. 30 minutes D. 60 minutes

Answer: A

11. Why is adrenaline allowed only 2 minutes before aerosol challenge, while mepyramine is allowed 30–45 minutes?

A. Adrenaline is rapidly absorbed and short-acting; mepyramine requires systemic distribution to block receptors B. Adrenaline acts only after 1 hour C. Mepyramine is destroyed within 5 minutes D. Guinea pigs degrade antihistamines instantly

Answer: A

12. What is the first clinical behavioral sign observed when the unprotected guinea pig is exposed to aerosolized antigen?

A. Sudden death B. Face preening / scratching of the nose C. Immediate flaccid paralysis D. Profuse urination

Answer: B

13. The full chronological sequence of symptoms in an unprotected guinea pig during anaphylactic challenge is:

A. Coma -> hyperactivity -> cough -> recovery B. Face preening -> restlessness -> coughing/sneezing -> dyspnea -> asphyxial convulsions -> collapse and death C. Tail erection -> salivation -> diarrhea -> death D. Sedation -> muscle relaxation -> righting reflex loss

Answer: B

14. The respiratory distress in guinea pig anaphylaxis is primarily caused by:

A. Massive histamine-mediated bronchiolar constriction and laryngeal edema B. Cardiac arrest due to hyperkalemia C. Paralysis of the phrenic nerve D. Acute hypoglycemia

Answer: A

15. What type of hypersensitivity reaction is represented by acute systemic anaphylaxis?

A. Type I (IgE-mediated immediate hypersensitivity) B. Type II (Antibody-dependent cytotoxic hypersensitivity) C. Type III (Immune complex-mediated hypersensitivity) D. Type IV (Delayed-type cell-mediated hypersensitivity)

Answer: A

16. Which primary mediator released from mast cell granules causes rapid bronchoconstriction and vasodilation?

A. Renin B. Histamine C. Aldosterone D. Cortisol

Answer: B

17. Histamine produces bronchoconstriction through which receptor subtype?

A. H_1 receptor (coupled to G_q/11, increasing IP_3/DAG and intracellular Ca^2+) B. H_2 receptor (coupled to G_s, increasing cAMP) C. H_3 receptor D. H_4 receptor

Answer: A

18. Histamine produces gastric acid secretion through which receptor subtype?

A. H_1 receptor B. H_2 receptor (coupled to G_s, increasing cAMP in parietal cells) C. H_3 receptor D. H_4 receptor

Answer: B

19. In the protocol table on Page 136, which antihistamine is tested at 1 mg/kg i.p.?

A. Mepyramine B. Triprolennamine C. Diphenhydramine D. Promethazine

Answer: B

20. In the protocol table on Page 136, what is the dose and route for diphenhydramine?

A. 1 mg/kg s.c. B. 10 mg/kg i.p. C. 50 mg/kg p.o. D. 100 mg/kg i.v.

Answer: B

21. In the protocol table on Page 136, what xanthine derivative is tested at 40 mg/kg i.p.?

A. Caffeine B. Aminophylline C. Theobromine D. Pentoxifylline

Answer: B

22. How much time is allowed after aminophylline injection before aerosol challenge?

A. 2 minutes B. 10 minutes C. 30 minutes D. 60 minutes

Answer: B

23. What is the mechanism of bronchodilation produced by aminophylline?

A. Muscarinic receptor stimulation B. Inhibition of phosphodiesterase (PDE) enzymes, elevating intracellular cAMP C. Direct activation of alpha-1 receptors D. Release of histamine from mast cells

Answer: B

24. What parameter is counted during aerosol exposure to measure onset of respiratory distress?

A. Number of heartbeats B. Number of "squeezes" of the nebulizer bulb / duration of exposure required for distress C. Number of steps walked D. Volume of water consumed

Answer: B

25. What is the major post-mortem pathological finding in a guinea pig dying of acute anaphylaxis?

A. Severely shrunken, collapsed lungs B. Massively inflated, pale, emphysematous lungs that do not collapse upon opening the thorax C. Massive cerebral hemorrhage D. Acute hepatic necrosis

Answer: B

26. Why do the lungs remain hyperinflated during guinea pig anaphylactic shock?

A. Complete obstruction of the bronchioles traps inspired air ("air trapping") B. The lungs fill completely with arterial blood C. Diaphragm is permanently contracted D. Trachea becomes ossified

Answer: A

27. What is the pharmacological relationship between adrenaline and histamine in anaphylaxis?

A. Competitive antagonism at H_1 receptors B. Physiological (functional) antagonism acting through opposing adrenergic receptors C. Chemical neutralization forming an inactive salt D. Pharmacokinetic metabolic antagonism

Answer: B

28. Which adrenergic action of adrenaline combats the severe hypotension and laryngeal edema of anaphylaxis?

A. Alpha-1 receptor-mediated vasoconstriction B. Beta-2 receptor-mediated vasodilation C. Alpha-2 receptor autoreceptor blockade D. Muscarinic M_2 stimulation

Answer: A

29. Which adrenergic action of adrenaline relieves the severe bronchospasm of anaphylaxis?

A. Alpha-1 vasoconstriction B. Beta-2 receptor-mediated bronchial smooth muscle relaxation C. Beta-1 chronotropic stimulation D. Nicotinic stimulation

Answer: B

30. In Question 2 on Page 136, why is adrenaline clinically superior to antihistamines in treating acute anaphylactic shock?

A. Antihistamines are toxic to the kidney B. Adrenaline rapidly reverses bronchospasm, supports blood pressure, and stabilizes mast cells, whereas antihistamines only block H_1 receptors C. Antihistamines take 24 hours to act D. Adrenaline is a competitive H_2 blocker

Answer: B

31. Can H_1 antihistamines completely prevent or reverse all features of anaphylaxis?

A. Yes, they cure 100% of symptoms B. No, because other mediators (leukotrienes/CysLTs, PAF, bradykinin) also contribute to shock and bronchospasm C. Yes, if given in extremely small doses D. No, because guinea pigs lack H_1 receptors

Answer: B

32. In Question 3 on Page 136, why is disodium cromoglycate USELESS once anaphylactic shock is established?

A. It is destroyed by gastric acid B. It acts only as a prophylactic mast cell stabilizer; once mast cells have degranulated, it cannot reverse mediator actions C. It stimulates histamine release from basophils D. It is a powerful bronchoconstrictor

Answer: B

33. When must disodium cromoglycate be administered to be therapeutically effective?

A. After the patient collapses B. Prophylactically before exposure to the provoking antigen C. Intravenously during cardiac arrest D. Only after adrenaline has failed

Answer: B

34. Which class of anti-inflammatory drugs is useful in the late-phase reaction of anaphylaxis to prevent recurrence?

A. Glucocorticoids (e.g. hydrocortisone, dexamethasone) B. Non-steroidal anti-inflammatory drugs (NSAIDs) C. Opioid analgesics D. Proton pump inhibitors

Answer: A

35. What is the effect of pre-treating a guinea pig with mepyramine before antigen aerosol?

A. Hastens the onset of death B. Significantly prolongs the incubation time before respiratory distress occurs and may prevent death C. Produces immediate hyperventilation D. Has zero effect on histamine-induced bronchospasm

Answer: B

36. Diphenhydramine belongs to which chemical class of H_1 antihistamines?

A. Ethanolamines B. Ethylenediamines C. Phenothiazines D. Piperazines

Answer: A

37. Promethazine belongs to which chemical class of H_1 antihistamines?

A. Alkylamines B. Phenothiazines C. Piperidines D. Phthalazinones

Answer: B

38. A common prominent side effect of first-generation H_1 antihistamines (like promethazine and diphenhydramine) is:

A. Severe insomnia B. Central nervous system sedation and drowsiness C. Acute hypertension D. Excessive salivation

Answer: B

39. Why do first-generation antihistamines cause prominent sedation while second-generation agents (e.g. cetirizine, loratadine) do not?

A. First-generation agents readily cross the blood-brain barrier and block central H_1 receptors B. Second-generation agents destroy histamine in blood C. First-generation agents activate GABA receptors directly D. Second-generation agents are administered only topically

Answer: A

40. The ultimate conclusion of Experiment 27 is that:

A. Histamine is not involved in anaphylactic reactions B. Adrenaline provides lifesaving physiological antagonism, while H_1 antihistamines provide competitive pharmacological protection against histamine-induced bronchospasm C. Antihistamines alone are sufficient to treat established anaphylactic collapse D. Disodium cromoglycate is the drug of choice for acute bronchospasm

Answer: B